

Payspin — Solution Architecture Document

For Technical Partner — Initial Design & Architecture Reference

Version: 1.0 | **Date:** May 2026 | **Author:** Moustafa — Founder, Payspin.io | **Status:** Final for Phase 1 kickoff

1. Product Overview

Payspin is a two-product fintech platform targeting Germany and the Netherlands, built on open banking and blockchain rails.

	Service 1 — Payspin P2P	Service 2 — Payspin Circles
What it is	Tikkie-style payment link service	ROSCA (rotating savings circle) service
User analogy	“Tikkie but for Germany and Netherlands”	“Digital Tandas / Geldkring / Sparkreis”
Primary market	DE + NL	DE + NL + EU diaspora communities
Core technology	Open banking (Yapily PISP)	Blockchain (Monerium + Circle + Gnosis Chain)
Launch phase	Phase 1	Phase 3

2. The Non-Custodial Principle — Read This First

Payspin never holds user funds at any point.

This is not a preference — it is the architectural foundation. It determines every technology decision in this document. Any design decision that results in Payspin holding, touching, or controlling user money must be escalated and approved before implementation.

- Payspin does not hold euros

- Payspin does not hold stablecoins
- Payspin does not operate a settlement account
- Payspin does not issue IBANs (Monerium does, in Phase 3)
- Payspin does not custody wallets (Circle MPC does, in Phase 3)

The regulatory consequence of this principle: Payspin operates as a software platform and commercial agent of licensed providers. No PI licence required to launch Phase 1.

3. Architecture — Service 1: P2P Payment Links

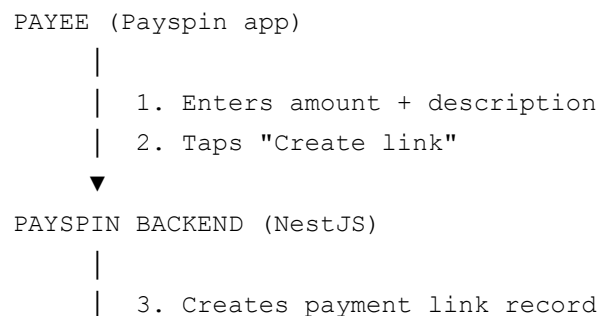
3.1 User Experience Reference — Tikkie (the model to replicate)

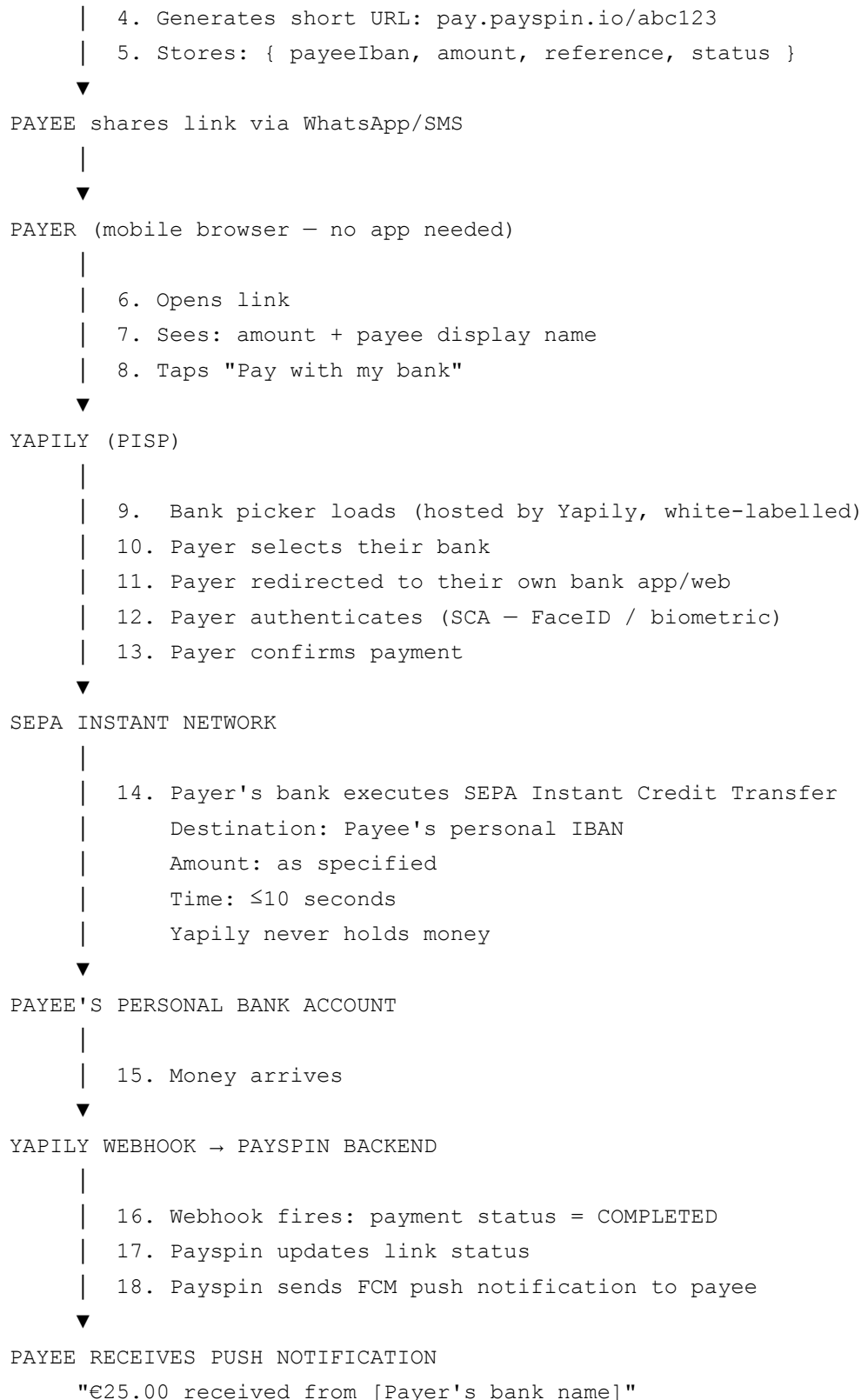
Tikkie (ING Bank, Netherlands) is the UX benchmark. Study the Tikkie app before designing anything. Key UX principles to copy:

- Payee creates a link in under 10 seconds
- Link is shared via WhatsApp / SMS / any messenger — one tap
- Payer opens link in mobile browser — **zero app install, zero registration**
- Payer selects their bank from a picker
- Payer authenticates with their own bank (FaceID / fingerprint)
- Payment confirmed instantly
- Payee receives a push notification

The critical difference from Tikkie: Payspin works across Germany AND Netherlands (Tikkie is NL-only), supports any EU bank (not just ING customers), and is open banking native (not iDEAL-only).

3.2 Architecture A1 — Open Banking, Open-Loop





Money path: Payer's Bank → SEPA Instant → Payee's Personal Bank **Hops:** 1 **Settlement:** ≤10 seconds **Payspin in money path:** Never

3.3 PISP Provider — Yapily

- **Licence:** Bank of Lithuania (LB002045) for EU / FCA for UK
 - **Coverage:** 2,000+ banks across 19 EU countries
 - **DE banks confirmed:** Deutsche Bank, Commerzbank, DKB, ING-DiBa, N26, Sparkasse network, Volksbanken network
 - **NL banks confirmed:** ING, ABN AMRO, Rabobank, Knab, ASN Bank, SNS Bank
 - **Payspin licence required:** None — operates under Yapily's PI licence
 - **Product used:** Yapily Connect (no-licence model) → upgrade to direct API after PI licence obtained
 - **Pricing (quoted):** €250/month platform + €500 per 1,500 transactions (~€0.33/tx)
 - **Key commercial question still open:** Confirm beneficiary IBAN is a per-transaction parameter (open-loop confirmation pending)
 - **Sandbox:** Available at yapily.com — start here before production integration
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4. Architecture — Service 2: Payspin Circles (ROSCA)

4.1 What a ROSCA is

A ROSCA (Rotating Savings and Credit Association) is a group savings mechanism:

- N members each contribute a fixed amount per cycle (e.g. €100/month)
- Each cycle, one member receives the full pot ($N \times €100$)
- The rotation continues until every member has received the pot once
- Example: 10 members $\times$ €100/month = €1,000 pot. Month 1 → Member 1. Month 2 → Member 2. Etc.

Known as: Geldkring (NL), Sparkreis (DE), Tanda (Mexico), Hui (China), Chit (India).

4.2 Architecture B — PISP + Monerium + Blockchain

PHASE A — PAYEE ONBOARDING (one-time)

PAYEE (Payspin app)

- |
- | 1. Registers on Payspin (Firebase Auth)
- | 2. Taps "Connect bank for Circles"

▼
MONERIUM OAUTH FLOW

- |
- | 3. Monerium OAuth popup opens (hosted by Monerium)
- | 4. Payee completes KYC: name, ID, address (2-5 min)
- | 5. Monerium creates:
 - | - Personal virtual IBAN (DE/NL format)
 - | - EVM wallet address on Gnosis Chain
 - | - Links IBAN ↔ wallet address

▼
PAYSPIN BACKEND

- |
- | 6. Receives Monerium OAuth callback
- | 7. Stores: { userId, moneyiumIban, walletAddress }
- | 8. Payee is now "Circles-enabled"

PHASE B – CONTRIBUTION (monthly, from each member)

CIRCLE MEMBER (Payspin app)

- |
- | 1. App shows "Your contribution is due: €100"
- | 2. Member taps "Pay now"

▼
PAYSPIN BACKEND

- |
- | 3. Creates payment request via Yapily API
 - | Beneficiary IBAN: Smart contract's Monerium IBAN
 - | Amount: €100

▼
YAPILY (PISP)

- |
- | 4. Bank picker loads
- | 5. Member selects their bank, authenticates
- | 6. SEPA Instant: Member's Bank → Circle Smart Contract IBAN

▼
MONERIUM

- |
- | 7. Receives SEPA credit at Circle's IBAN
- | 8. AML check passes
- | 9. Mints EURE on Gnosis Chain
- | 10. EURE sent to ROSCA Smart Contract address

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GNOSIS CHAIN – ROSCA SMART CONTRACT

- |
- | 11. Contract records: member X contributed €100

- | 12. Checks if all members have contributed this round
 - | 13. If yes → triggers payout to this round's recipient
-

PHASE C – PAYOUT (to recipient of this round)

ROSCA SMART CONTRACT

- |
- | 1. Round complete – all members contributed
- | 2. Contract calls: `transfer(recipientWallet, fullPot)`
- | 3. EURE transferred to recipient's EVM wallet (≤5 seconds)



RECIPIENT (Payspin app)

- |
- | 4. EURE arrives in wallet
- | 5. Choice A: Keep as EURE (earn yield on Aave V3)
- | 6. Choice B: Burn to fiat – tap "Withdraw to bank"



MONERIUM (burn flow)

- |
- | 7. Recipient initiates burn via Monerium API
- | 8. EURE burned on-chain
- | 9. Monerium initiates SEPA bank transfer to recipient's personal IBAN (the real bank account)
- | 10. Time: ≤10 seconds via SEPA Instant



RECIPIENT'S PERSONAL BANK ACCOUNT

4.3 Smart Contract — ROSCA Logic

The smart contract handles all group mechanics on-chain. Key functions:

`contribute(circleId, memberId)`

- Accepts EURE from member
- Marks member as paid for this round
- Checks if all members paid → triggers payout

`claimPayout(circleId)`

- Callable by current round recipient only
- Transfers full pot to recipient wallet

`defaultMember(circleId, memberId)`

- Admin/multisig only
- Handles non-paying member – proportional refund to others

```
getCircleState(circleId)
```

→ Returns: members, round number, paid status, recipient

```
pauseCircle(circleId)
```

→ Emergency pause — 3-of-5 multisig required

Smart contract chain: **Gnosis Chain** (low gas ~\$0.001/tx, EURE native) Automation: **Gelato Web3 Functions** (cron-based poke() for missed contributions) Audit: Mandatory before mainnet — budget €30,000–60,000

4.4 Blockchain Partners

Partner	Role	Licence	What Payspin does
Monerium	Fiat ↔ EURE bridge. Issues virtual IBANs. KYC on payees.	EMI (Iceland CBI) + MiCA Art.48	OAuth integration + webhook listener
Circle (MPC Wallet)	Non-custodial wallet infrastructure. Gas Station for gasless tx.	N/A — infrastructure	SDK integration for wallet creation
Gnosis Chain	L2 blockchain network. EURE native. ~\$0.001 gas.	N/A — public chain	Deploy smart contracts here
Aave V3 (Gnosis)	Yield on idle ROSCA funds (~4% APY)	N/A — DeFi protocol	Optional Phase 3+

5. Full Technology Stack

5.1 Core Stack Decision

Layer	Technology	Why
Mobile app	React Native (Expo)	Single codebase for iOS + Android. Fast iteration.
Payer web page	Next.js 14 App Router	Server-side rendered, fast load, edge runtime
Backend API	NestJS (Node 20, TypeScript)	Modular, decorator-based, production-proven for fintech

Database	PostgreSQL 16	Transactions, ACID compliance, IBAN encryption at rest
Auth	Firebase Auth	Email, Google, Apple login. JWT tokens.
Push notifications	Firebase FCM	Free, reliable, cross-platform
Job queue	BullMQ + Redis	Async webhook processing, retry logic
Smart contracts	Solidity 0.8.24 + Foundry	Industry standard, Foundry for testing
Blockchain	Gnosis Chain	EUR native, low gas, EVM compatible
Cloud	GCP Cloud Run (europe-west1)	Serverless, auto-scale, Frankfurt region
Repo	Turborepo monorepo	Shared types, unified CI/CD

5.2 Monorepo Structure

```

payspin/
├── apps/
│   ├── api/           ← NestJS backend (Cloud Run)
│   ├── web/           ← Next.js payer landing pages (Vercel)
│   └── mobile/        ← React Native Expo (App Store + Play Store)
├── packages/
│   ├── shared-types/  ← TypeScript interfaces used across all apps
│   └── pisp-provider/  ← Yapily abstraction (swap providers easily)
├── contracts/
│   ├── src/           ← Solidity smart contracts
│   └── test/          ← Foundry tests (invariant + fuzz)
├── infrastructure/
│   └── sql/           ← Database migrations
└── turbo.json

```

5.3 Key Architectural Abstractions

PisProvider interface — The most important abstraction in the codebase. All PISP calls go through this interface. Changing from Yapily to IbanXS or Token.io is a config change, not a code change.

```
interface PisProvider {
```

```

initiatePayment(params: {
  amountCents: number;
  currency: string;
  beneficiaryIban: string;    // ← This is the open-loop key
  beneficiaryName: string;
  reference: string;
  redirectUri: string;
  paymentLinkId: string;
}): Promise<{ tokenId: string; redirectUrl: string }>;

getPaymentStatus(tokenId: string): Promise<PaymentStatus>;
verifyWebhookSignature(payload: string, sig: string): boolean;
}

```

MoneyPath interface — Used in Phase 3. Routes between fiat (Yapily only) and blockchain (Yapily + Monerium) based on the payee's account type.

6. Database Schema — Core Tables

```

-- Users (payees registered on Payspin)
users (id, firebase_uid, email, display_name, created_at)

-- Bank accounts (payee's personal IBAN — AES-256 encrypted at rest)
bank_accounts (id, user_id, iban_encrypted, iban_last4,
               account_holder, bank_name, verified)

-- Payment links
payment_links (id, short_code, payee_user_id, bank_account_id,
               amount_cents, currency, description, status,
               link_type, max_uses, use_count, expires_at)
-- status: ACTIVE | EXPIRED | CANCELLED | SETTLED | COLLECTING
-- link_type: SINGLE (P2P) | MULTI (Verein, events)

-- Payments (one per payer per link)
payments (id, payment_link_id, yapily_payment_id,
          amount_cents, status, payer_bank_name,
          initiated_at, completed_at, webhook_raw)

-- Webhook idempotency (prevents double-processing)
webhook_events (id, event_type, processed_at, payload)

-- Phase 3: ROSCA circles
circles (id, name, host_user_id, member_count,
         contribution_cents, cycle_duration_days,

```

```
smart_contract_address, status, current_round)
```

```
circle_members (id, circle_id, user_id, monerium_iban,  
                wallet_address, payout_order, status)
```

7. Security Requirements (Non-Negotiable)

Every item below must be implemented before production:

Requirement	Implementation
IBAN encryption at rest	AES-256-GCM, per-record IV, key from env var
Webhook signature verification	HMAC-SHA256, timing-safe comparison
Webhook idempotency	webhook_events table, check before processing
Short code unpredictability	crypto.randomBytes(), not sequential
IBAN never in API response to payer	Only server-side, never in browser
No PII in logs	Never log IBAN, full reference, or user data
Firebase token expiry	1 hour max, refresh on resume
Rate limiting	10 req/min per IP on public endpoints
AML velocity checks	>10 payments/24h or >€5,000/24h → compliance flag

8. Third-Party Integrations Summary

Yapily (Phase 1, 2, 3)

- **What:** PISP payment initiation
- **Docs:** docs.yapily.com
- **Auth:** OAuth2 client credentials
- **Key endpoint:** POST /payments (with beneficiary IBAN per request)
- **Webhook:** X-Yapily-Signature HMAC header

- **Sandbox:** Available immediately — start here
- **Contact:** Katie Hayton (via current thread) for production access

Monerium (Phase 3)

- **What:** EMI bridge. Issues virtual IBANs. Mints/burns EURE on Gnosis Chain.
- **Docs:** monerium.dev
- **Auth:** OAuth2 (hosted KYC flow for end users)
- **Key flow:** User completes Monerium OAuth → receives IBAN + wallet address
- **Webhook:** order.processed event = EURE minted
- **Sandbox:** monerium.dev/sandbox

Circle (Phase 3)

- **What:** MPC wallet infrastructure. Gas Station for gasless transactions.
- **Docs:** developers.circle.com
- **Auth:** API key
- **Key product:** Programmable Wallets + Gas Station
- **Sandbox:** Available immediately

Firebase (Phase 1, 2, 3)

- **What:** Auth, Firestore, FCM push notifications
- **Docs:** firebase.google.com/docs
- **Auth:** Service account JSON
- **Free tier:** 10k MAU, 50k Firestore reads/day, FCM free

9. Phased Delivery Plan

Phase 1 — P2P MVP (Target: 8 weeks to production)

Goal: Working Tikkie-equivalent app in Germany and Netherlands, live on App Store and Play Store.

Scope:

- Payee registers, adds personal IBAN
- Creates payment link (fixed or variable amount)
- Shares link via native share sheet / WhatsApp
- Payer pays in browser (zero install)
- Yapily PISP integration (payment initiation)
- Webhook handler (payment confirmation)
- FCM push notification to payee
- Payment history screen
- Link management (view, cancel, expiry)

Out of scope for Phase 1:

- Blockchain / EURE / Monerium
- ROSCA circles
- Yield
- Business features (DATEV export, invoicing)
- Cross-border beyond DE + NL

Week-by-week:

Week	Deliverable
1	Monorepo scaffold, Firebase setup, Postgres schema, API auth guard
2	IBAN storage (encrypted), payment links CRUD, short URL generation
3	Yapily sandbox integration — payment initiation + redirect flow
4	Webhook handler + idempotency + AML velocity checks
5	Next.js payer landing page + bank picker + callback handler
6	React Native app — auth, IBAN onboarding, link creation, share sheet
7	FCM push, payment history, end-to-end integration testing (10 scenarios)

First euro on mainnet target: Day 42–45

Phase 2 — P2P Enhancement (Target: Weeks 9–16)

Goal: Polish the P2P experience, improve conversion, add business value features.

Scope:

- Push notification improvements (real-time balance, history)
 - Link analytics (how many people opened, how many paid)
 - Multi-use links (Verein dues, event tickets)
 - QR code generation for in-person use
 - DATEV-compatible CSV export (critical for German freelancers)
 - Recurring payment links (retainer, tutor, cleaner)
 - IBAN verification via Yapily AIS (optional step at registration)
 - Variable Recurring Payments (VRP) for scheduled contributions (Yapily supports this)
 - Language: German and Dutch full localisation
 - Upgrade to Yapily direct API (after PI licence — replaces Connect tier)
-

Phase 3 — Payspin Circles / ROSCA (Target: Weeks 17–32)

Goal: Launch ROSCA product on blockchain. Monerium + Circle + Gnosis Chain.

Scope:

- Monerium OAuth integration (payee KYC + virtual IBAN issuance)
- Circle MPC wallet creation per user
- ROSCA smart contract (Solidity) — contribute, claim, default, pause
- Gelato automation for round management
- Smart contract audit (mandatory pre-mainnet — budget €30,000–60,000)
- EURE contribution flow (Yapily initiates SEPA to Monerium IBAN)

- On-chain payout to recipient wallet
- Fiat off-ramp via Monerium burn → SEPA
- Optional: Aave V3 yield on idle circle funds (MiCA opt-in required)
- Optional: Emergency multisig (3-of-5 Safe) for contract pause

10. Compliance Architecture

Payspin's non-custodial design means a minimal compliance footprint. Here is what is required and what is delegated:

Requirement	Who handles it	How
KYC on payers	Not required	Payers never register on Payspin
KYC on payees (Phase 1)	Payspin	Light: email + IBAN verification
KYC on payees (Phase 3)	Monerium	Full KYC via OAuth flow (delegated)
AML monitoring	Payspin + Yapily	Velocity checks in backend + Yapily's own AML
SEPA AML	Payer's bank	Banks do their own AML on debited accounts
MiCA compliance (EURE)	Monerium	EURE is MiCA Art.48 compliant EMT
GDPR	Payspin	IBAN encrypted, minimal data retention
Travel Rule (>€1,000)	Yapily + Monerium	Both providers handle Travel Rule metadata

AML minimum implementation (Phase 1):

- Payee velocity: >10 distinct payers in 24h → compliance_flags table + alert
- Amount velocity: >€5,000 received in 24h → flag
- IP rate limit: >20 payment attempts from same IP in 1h → block + flag

11. Legal Entity Structure

Entity	Type	Purpose	Status
Payspin B.V.	Dutch B.V.	Operating company. Contracts with Yapily, Monerium, Circle. Employs team.	Incorporate immediately via Firm24.nl
Payspin UAB	Lithuanian UAB	Regulated entity. Holds PI licence from Bank of Lithuania.	Incorporate in Month 2, apply for licence Month 3

PI Licence (Lithuania):

- Regulator: Bank of Lithuania
- Licence type: Payment Initiation Services only (PSD2 Annex I point 7)
- Capital required: €50,000
- Application fee: €898
- Timeline: 3–6 months
- Unlocks: Direct Yapily API (replacing Connect tier), Token.io, Enable Banking

12. Open Items — Decisions Pending

The following must be resolved before or during Phase 1 development:

Item	Owner	Deadline	Impact if delayed
Yapily open-loop confirmation (beneficiary IBAN per tx)	Moustafa / Katie	This week	Blocks all Phase 1 architecture
Yapily pricing negotiation	Moustafa	Before signing	Cost structure for unit economics
Payspin B.V. incorporation	Moustafa	This week	Required for Yapily due diligence form
Bunq Business bank account	Moustafa	Week 2	Required for Yapily form
AML policy document (1 page)	Moustafa	Week 2	Required for Yapily

	+ Tech		onboarding
Smart contract audit firm selection	Tech	Phase 3 start	Mandatory before mainnet Circles
Monerium developer account	Tech	Phase 3 start	Required for ROSCA architecture
Circle developer account	Tech	Phase 3 start	Required for MPC wallets

13. Reference Links

Resource	URL
Yapily API docs	docs.yapily.com
Yapily sandbox signup	yapily.com/pricing
Monerium developer docs	monerium.dev
Circle developer docs	developers.circle.com
Gnosis Chain docs	docs.gnosischain.com
Gelato automation	docs.gelato.network
Firebase docs	firebase.google.com/docs
GCP Cloud Run	cloud.google.com/run
Foundry (smart contract testing)	book.getfoundry.sh
OpenZeppelin contracts	docs.openzeppelin.com
Bank of Lithuania PI licence	lb.lt/en/licences
Firm24 NL B.V. incorporation	firm24.nl

